Experiment No:1.a
Analyze the trend of data science job postings over The last decade

AIM:

To Analyze the trend of data science job postings over the last Decade

DESCRIPTION:

Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Year': list(range(2015, 2026)),
    'Job Postings': [180, 320, 480, 620, 850, 1250, 1650, 2150,
2800, 3500, 4300]
}

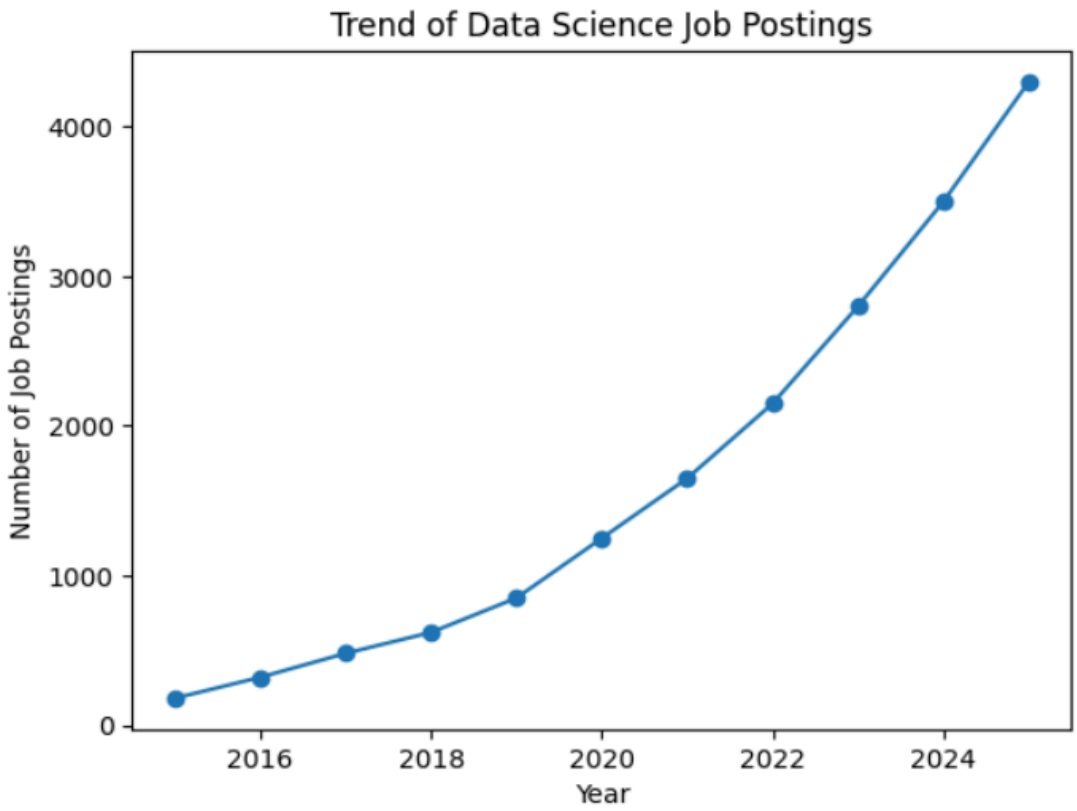
df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')

plt.show()
```

OUTPUT:



RESULT:

The graph shows that data science job posting increases from 2015 to 2026 .The highest number of job postings was in 2026

Experiment No:1.b

Analyze and visualize the distribution of various data science roles (DataAnalyst, Data Engineer, Data Scientist, etc.) from a dataset.

AIM:

To Analyze and visualize the distribution of various data science roles from a dataset.

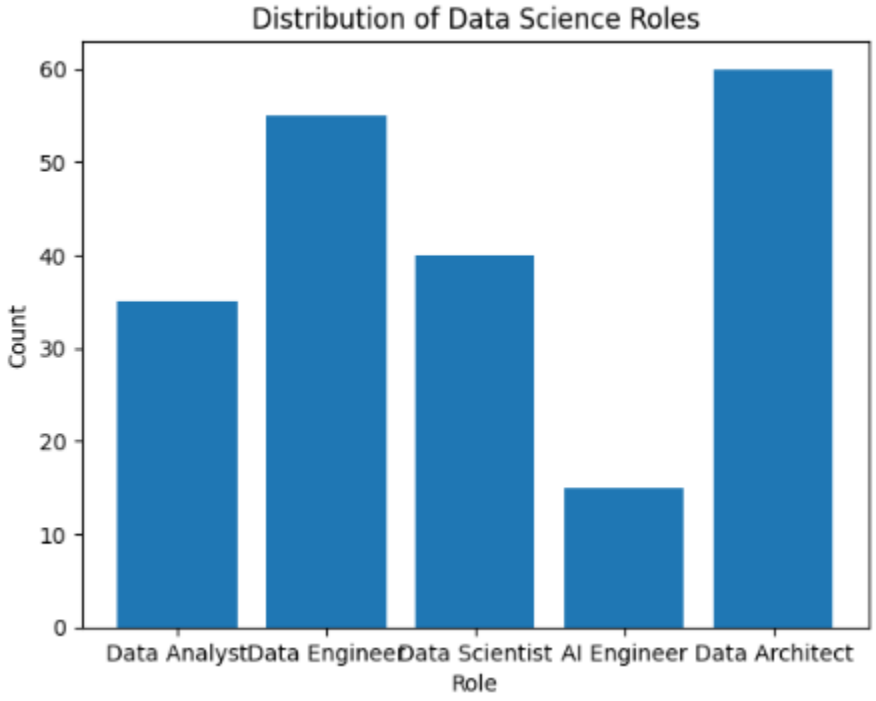
DESCRIPTION:

Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist',
'AI Engineer',
'Data Architect']
counts = [35, 55, 40, 15, 60]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

OUTPUT:



RESULT:

The bar chart shows that Data Architect has the highest count (59), followed by Data Engineer (55), Data Scientist (40), Data Analyst (35), and AI Engineer (15). Therefore, Data Architect is the most common role, while AI Engineer is the least common.

Experiment No:1.c
Conduct an experiment to differentiate Structured , Un-structured and Semistructured data based on data sets given.

AIM:

To Conduct an experiment to differentiate Structured , Un-structured and Semistructured data based on data sets given.

DESCRIPTION:

Create small datasets for each type and explain their characteristics.

CODE:

```
structured_data = pd.DataFrame({
    'ID': [101, 102, 103],
    'Name': ['David', 'Emma', 'Frank'],
    'Age': [22, 28, 40]
})
print("Structured Data:\n", structured_data)

unstructured_data = "Machine learning helps computers learn from data."
print("\nUnstructured Data:\n", unstructured_data)

semi_structured_data = {
    'ID': 101,
    'Name': 'David',
    'Attributes': {
        'Height': 178,
        'Weight': 75
    }
```

```
}
print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

```
Structured Data:
     ID   Name  Age
0  101  David   22
1  102   Emma   28
2  103  Frank   40

Unstructured Data:
 Machine learning helps computers learn from data.

Semi-structured Data:
 {'ID': 101, 'Name': 'David', 'Attributes': {'Height': 178, 'Weight': 75}}
```

[

RESULT:

Structured data is the easiest to query via sql

semi structured is self describing but schema flexible.

unstructured data carries no explicit schema

Experiment No:1.d
Conduct an experiment to encrypt and decrypt given sensitive data.

AIM:

To Conduct an experiment to encrypt and decrypt given sensitive data.

DESCRIPTION:

Use the cryptography library to encrypt and decrypt a piece of data.

CODE:

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"")
token
b'...'
f.decrypt(token)
b'fundamental of data science'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"fundamental of data science"
cipher_text = cipher_suite.encrypt(plain_text)
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

OUTPUT:

```
Original Data: b'fundamental of data science'
Encrypted Data:
b'gAAAAABqhGX8OLUp-yzVPjwSzPanb9xYdER9UBcN1fPXXBZV1SOd0Rn-ndBY
mx-SNEWz6AgMvMl0tG8AOiBPlUVVSF_odKDiZklvH894Tv0RwOvwP2coDR8='
Decrypted Data: b'fundamental of data science'
```

RESULT:

The fernet Symmetric -key Algorithm successfully encrypted back original data text,verifying both any tumering with the cipherset would cause decryption to fail.